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| STUDENT NAME: | Francisco Sandoval Hernández |
| SUPERVISOR NAME: | Moisés Martínez Muñoz |

TUTORING SESSION Nº:

☐ 1. Start
☐ 2. State of the Art / Previous Research
☒ 3. Project Plan
☐ 4. Project Development
☐ 5. Technical Work Closure
☐ 6. Report Review
☐ 7. Other: Specify

DATE: 06/02/2026
START: 19:00
END: 20:00

**DISCUSSED TOPICS:**

During this tutoring session, the practical project plan was reviewed after the first exploratory experiments. The original route based on hand detection and static handshape recognition was considered useful as an initial feasibility study, but it was agreed that the main technical development should move towards landmark-based word-level recognition.

WLASL30 was selected as the first controlled benchmark. This subset was considered appropriate because it allowed the student to test several temporal architectures with limited local hardware, while keeping the classification task difficult enough to represent dynamic sign recognition. The supervisor and student agreed that increasing the vocabulary size should be left for later phases, once a stable baseline had been obtained.

The expected recognition pipeline was defined as video acquisition, frame decoding, landmark extraction with MediaPipe, landmark normalization, sequence modelling and metric-based evaluation. This pipeline would allow raw ASL videos to be transformed into structured sequences of hand keypoints, which could then be processed by recurrent or attention-based models.

Different modelling options were discussed. CNN-LSTM and LSTM-based models were considered, but GRU-based alternatives were also included because they could provide a good balance between temporal modelling capacity and computational cost. Transformer-based models were considered as a comparison baseline, although their size and data requirements were identified as possible limitations.

The evaluation strategy was also discussed. It was agreed that simple accuracy would not be enough to understand the behavior of the models. The experiments should include Top-1 accuracy, Top-5 accuracy, Macro-F1 and confusion matrices, so that the student could analyze both general performance and specific class-level errors.

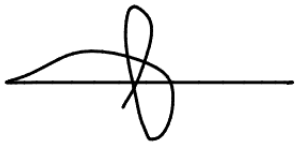
**ACUERDOS PARA LA SIGUIENTE REUNIÓN:**

| Item | Agreement / Task | Person in Charge | Deadline |
|---|---|---|---|
| **1** | Build the WLASL30 landmark dataset and verify that the extraction process works correctly. | F.S.H | 20/02/2026 |
| **2** | Implement a first temporal baseline for word-level recognition. | F.S.H | 20/02/2026 |

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| **3** | Compare several model families under the same data split and metrics. | F.S.H | 20/02/2026 |
| **4** | Define the evaluation metrics to use in the experiments: Top-1, Top-5, Macro-F1 and confusion matrices | F.S.H | 20/02/2026 |

**SCHEDULED DATE OF THE NEXT MEETING:** **20/02/2026 – 19:00**

**Supervisor's Comments:**

| Supervisor's Signature: | Student's Signature: |
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|  Moisés Martínez Muñoz |  |